Melanoma

Staged by **Breslow thickness + ulceration**; **LDH is a staging modifier** in M1 disease · ~50% of cutaneous melanomas carry **BRAF V600** · **immunotherapy sequencing** is the main treatment question

*Stanford Oncology Handbook 2025, Melanoma — an **advanced-disease** chapter. Biopsy technique, excision margins, SLNB indications, dosing, surveillance and the variant/toxicity material are **not in it**; those are marked † and drawn from NCCN and product labeling.*

WORKUP

- † **Excisional biopsy** with 1–3 mm margins, full thickness, oriented along lymphatic drainage — **avoid a superficial shave**, which compromises Breslow depth
- Pathologic confirmation; report **Breslow thickness and ulceration** († mitotic rate is still reported though no longer a staging criterion)
- Imaging: CT chest/abdomen/pelvis (add neck if head-and-neck primary), or PET/CT
- **MRI brain for stage III–IV**
- Labs: CBC, CMP, **serum LDH**
- **BRAF V600 testing in all advanced cutaneous melanoma**
- If BRAF wild-type → **NGS**; in acral/mucosal look specifically for **KIT mutations or TRK fusions**

STAGING — AJCC 8TH

T by thickness	T1 ≤1 mm · T2 >1–2 · T3 >2–4 · T4 >4 mm; suffix **a = no ulceration, b = ulcerated**
T1a vs T1b	**T1a = <0.8 mm without ulceration**; T1b = <0.8 mm ulcerated, or 0.8–1.0 mm either way
N	N1 = 1 node or in-transit/satellite alone · N2 = 2–3 · **N3 = ≥4, or matted**
M	M1a skin/soft tissue/non-regional node · M1b lung · M1c other visceral · **M1d = CNS**
LDH	Explicit M-category modifier: **(0) = normal, (1) = elevated**

† LOCALIZED MANAGEMENT (NCCN)

Breslow depth	Excision margin
In situ	0.5–1.0 cm
≤1.0 mm	1 cm
>1.0–2.0 mm	1–2 cm
>2.0 mm	2 cm

† **Sentinel lymph node biopsy:** not recommended for **T1a** (risk <5%) · discuss for **T1b** · recommended for **T2 and above**. **Completion lymph node dissection is no longer routine** after a positive sentinel node (MSLT-II, DeCOG-SLT) — nodal ultrasound surveillance instead.

TREATMENT PATHWAY

Localized and resectable

Surgery is potentially curative
wide local excision by Breslow depth
† SLNB for T1b and above

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Adjuvant anti-PD-1
nivolumab or pembrolizumab × 1 year
for **resected stage IIB and higher**
† BRAF V600: dabrafenib + trametinib an option

Resectable stage III — neoadjuvant preferred

Give immunotherapy before surgery
NADINA: ipilimumab 1 + nivolumab 3 mg/kg ("flipped dose") ×2 pre-op, then adjuvant nivo
S1801: pembrolizumab 200 mg ×3 pre-op
improves EFS vs adjuvant therapy alone

Unresectable stage III or IV

First line — immunotherapy
nivolumab + ipilimumab (CheckMate-067)
nivolumab + relatlimab (RELATIVITY-047)
anti-PD-1 alone if not fit for a combination

↓

BRAF V600-mutant sequencing
immunotherapy FIRST, then BRAF/MEK
DREAMseq 2-yr OS **72% vs 52%**
exception: visceral crisis, or steroid-dependent
brain mets → brief BRAF/MEK then switch (sandwich)

↓

Later lines
TIL therapy after ICI (RR ~30%)
T-VEC for cutaneous/subcutaneous/nodal disease
chemotherapy last resort — RR <20%

↓

Oligometastatic
metastasectomy **not preferred alone** if fit
for systemic therapy — give adjuvant therapy
after complete resection

Brain metastases

Steroid requirement is the discriminator
asymptomatic, no steroids: **ipi/nivo**
intracranial ORR 53–57% (CheckMate-204)
symptomatic or on steroids: RR only ~17%
→ add surgery + targeted therapy; SRS, not WBRT

REGIMENS & DOSING

Regimen	Dosing	Setting
Pembrolizumab	† 200 mg q3wk or 400 mg q6wk	adjuvant ×1 yr; metastatic
Nivolumab	† 240 mg q2wk or 480 mg q4wk	adjuvant ×1 yr; metastatic
Ipilimumab + nivolumab	† ipilimumab 3 mg/kg + nivolumab 1 mg/kg q3wk ×4, then nivolumab maintenance	1L metastatic (CheckMate-067)
— neoadjuvant "flipped"	**ipilimumab 1 mg/kg + nivolumab 3 mg/kg** × 2 cycles pre-op	resectable stage III (NADINA)
Nivolumab + relatlimab	† 480 mg + 160 mg q4wk (Opdualag)	1L metastatic
Dabrafenib + trametinib	† 150 mg BID + 2 mg daily	BRAF V600; also adjuvant
Encorafenib + binimetinib	† 450 mg daily + 45 mg BID	BRAF V600
Vemurafenib + cobimetinib	† 960 mg BID + 60 mg daily (21/7)	BRAF V600
Imatinib	† for **activating** KIT mutations	acral / mucosal
TIL therapy (lifileucel)	resect → expand *ex vivo* with IL-2 → lymphodepleting chemo → single infusion → high-dose IL-2	PD-1-refractory
T-VEC	intralesional oncolytic HSV expressing GM-CSF	cutaneous, subcutaneous, nodal
High-dose IL-2	largely replaced by checkpoint and targeted therapy	historical
Chemotherapy	dacarbazine · temozolomide · carboplatin + paclitaxel	salvage; RR <20%, 4–6 months

† Doses are from product labeling — the handbook chapter states dosing only for the two neoadjuvant regimens. Cytotoxic chemotherapy "no longer plays a well-defined role" in metastatic melanoma.

BIOMARKER → THERAPY

Alteration	Therapy
BRAF V600 (~50%)	BRAF + MEK doublet — always dual, never single-agent BRAF (resistance, and BRAFi skin toxicity). Immunotherapy still works and generally comes **first**
KIT (~15% of acral/ mucosal)	Imatinib — responses with **activating mutations** (~1/3 respond); **KIT amplification alone typically does not respond**
TRK fusion	Found on NGS in BRAF wild-type acral/mucosal disease
† HLA-A*02:01	**Tebentafusp** in metastatic **uveal** melanoma; also eligibility for brenetafusp trials in cutaneous disease
BRAF wild-type	Send NGS for other actionable drivers

Melanoma

Page 2 — trial evidence, staging detail, immune toxicity and variants

TRIAL EVIDENCE

CheckMate-067 (10-yr)	ipi/nivo vs nivo vs ipi — median OS **72** / **37** / **20 months**; 10-yr OS **43%** / **37%** / **19%**. **If progression-free at 3 years, 10-yr melanoma-specific survival >96%** (Wolchok, NEJM 2024)
CheckMate-204	brain metastases on ipi/nivo — asymptomatic: intracranial ORR 53–57%, 36-mo iPFS ~54%, OS ~72%. Symptomatic or on steroids: RR ~17%, iPFS ~19%, OS ~37% (Long, JCO 2023)
RELATIVITY-047	relatlimab + nivolumab vs nivolumab — median PFS **10.1 vs 4.6 months** (Tawbi, NEJM 2022)
NADINA	neoadjuvant ipi/nivo → adjuvant nivolumab — EFS and distant-metastasis-free survival benefit; **major pathologic response 58%**; grade 3+ AEs ~30% (Rozeman, Nat Med 2024)
SWOG S1801	neoadjuvant → adjuvant pembrolizumab — 2-yr EFS **72% vs 48%** with adjuvant alone; no added surgical complications (Patel, NEJM 2023)
DREAMseq	BRAF V600: IO first then targeted vs the reverse — 2-yr OS **72% vs 52%** (Atkins, JCO 2023)
SECOMBIT	at 4 years, IO-first or sandwich beats targeted-first for PFS and OS (Ascierto, JCO 2023)
Dabrafenib + trametinib	pooled phase III: **34% alive at 5 years**, ~20% progression-free; better with normal LDH, good performance status, <3 metastatic sites
TIL therapy	RR ~30% with durable remissions; superior to ipilimumab in PD-1-refractory disease (Warner, Clin Cancer Res 2022)

† ADJUVANT TRIALS

KEYNOTE-716	pembrolizumab, resected **stage IIB/IIC** — 3-yr RFS **76.2% vs 63.4%** (HR 0.65)
CheckMate 76K	nivolumab 480 mg q4wk × 12 months, resected **stage IIB/IIC** — RFS HR **0.42**, DMFS HR 0.47
KEYNOTE-054	pembrolizumab, resected **stage III**
CheckMate 238	nivolumab vs ipilimumab, resected stage IIIB/C–IV
COMBI-AD	dabrafenib + trametinib × 12 months, resected stage III **BRAF V600**

All are 1-year regimens. **The stage IIB/IIC trials have shown RFS and DMFS benefit but no definitive OS benefit yet** — worth saying out loud when consenting a patient with resected node-negative disease.

STAGE GROUPINGS

Clinical	0 Tis · IA T1a · IB T1b or T2a · IIA T2b/T3a · IIB T3b/T4a · IIC T4b · **III = any N ≥N1, M0** · IV = M1
Pathologic IIIA	T1a–T2a with N1a or N2a
Pathologic IIIB	T0 with N1b/c · T1a–T2a with N1b/c or N2b · T2b/T3a with N1a–c or N2a/b
Pathologic IIIC	T0 with N2b/c or N3b/c · T1a–T3a with N2c or N3a–c · T3b/T4a with any N ≥N1 · T4b with N1a–N2c
Pathologic IIID	**T4b with N3a/b/c**
N suffixes	**a** = clinically occult (found on SLNB) · **b** = clinically detected · **c** = in-transit, satellite or microsatellite disease present

† IMMUNE-RELATED ADVERSE EVENTS

- **Grade 1:** usually continue with close monitoring. **Grade 2:** hold the checkpoint inhibitor; prednisone 0.5–1 mg/kg/day if persistent
- **Grade 3–4:** hold or permanently discontinue; **prednisone or methylprednisolone 1–2 mg/kg/day**, tapered over at least 4–6 weeks
- **Steroid-refractory colitis** → infliximab or vedolizumab. **Hepatitis** → mycophenolate, **not infliximab** (hepatotoxicity)
- **Myocarditis** → urgent high-dose steroids and permanent discontinuation — rare but highly lethal
- **Endocrine events are the exception:** hypophysitis, thyroiditis, adrenal insufficiency and type 1 diabetes are usually **irreversible**, but you replace the hormone and generally **continue** the checkpoint inhibitor
- Monitor TSH and free T4 at baseline and every 4–6 weeks; check HBsAg and HBcAb before starting
- Onset is typically weeks to months — **but can occur after discontinuation**. Ipilimumab-containing regimens carry the highest risk

VARIANTS & SPECIAL SITUATIONS

Acral (~2–3%)	More common in darker skin. Lower TMB, lower BRAF rate, **lower immunotherapy response**, more aggressive — check KIT
Mucosal (~1–2%)	Distinct genetics, low BRAF, poor immunotherapy response, presents late because the site is hidden; KIT actionable
Uveal	Biologically distinct; metastasizes to **liver first**. Localized: plaque radiation or enucleation. Metastatic: liver-directed therapy (HAI, TACE, radioembolization, ablation); † **tebentafusp if HLA-A*02:01**
† Desmoplastic	Usually BRAF wild-type with **very high TMB** — unusually responsive to anti-PD-1. SLNB value is debated in pure desmoplastic disease
† Lentigo maligna	Margin assessment is difficult (subclinical extension); imiquimod or radiation if not a surgical candidate
† In-transit disease	Isolated limb infusion or perfusion, T-VEC, intralesional therapy, or systemic therapy
† Unknown primary	Treat by stage as though cutaneous — outcomes are similar or better

† SURVEILLANCE (NCCN)

- **Stage IA–IIA:** H&P q6–12 months × 5 years, then annually; **no routine imaging**
- **Stage IIB–IV:** H&P q3–6 months × 2 years, then q3–12 months × 3 years; imaging q3–12 months for 2–3 years
- **Annual skin examination for life**, plus education on skin self-examination and sun protection
- **Pseudoprogression:** immunotherapy responses can be slow with transient worsening before shrinkage — especially with ipilimumab-containing regimens. † iRECIST requires confirmatory imaging before calling progression

† HEREDITARY PREDISPOSITION

CDKN2A / p16	Familial melanoma — also confers pancreatic cancer risk
BAP1	Uveal melanoma, mesothelioma, renal cell carcinoma, cutaneous melanocytic tumors
MC1R variants	Red hair and fair skin phenotype; modest risk modifier